Experiment 5: Develop a React application that demonstrates component composition and the use of props to pass data. Create two components: FigureList: A parent component responsible for rendering multiple child components. BasicFigure: A child component designed to display an image and its associated caption. Use the FigureList component to dynamically render multiple BasicFigure components. Pass image URLs and captions as props from the FigureList component to each BasicFigure component. Style the BasicFigure components to display the image and caption in an aesthetically pleasing manner. Arrange the BasicFigure components within the FigureList in a grid or list format. Allow users to add or remove images dynamically. Add hover effects or animations to the images for an interactive experience.

App.js

```
import React from 'react';
import './App.css';
import FigureList from './FigureList';

function App() {
  return (
    <div className="App">
      <h1>Figure Gallery</h1>
      <FigureList />
    </div>
  );
}

export default App;
```

FigureList.js

```
import React, { useState } from "react";

// BasicFigure Component (Child)
function BasicFigure({ image, caption, onRemove }) {
    return (
        <div className="basic-figure">
            <img src={image} alt={caption} className="figure-image" />
            <figcaption className="figure-caption">{caption}</figcaption>
            <button className="remove-button" onClick={onRemove}>
                Remove
            </button>
        </div>
    )
}

// FigureList Component (Parent)
function FigureList() {
    const [figures, setFigures] = useState([
        { id: 1, image: "https://www.w3schools.com/w3images/lights.jpg", caption: "Beautiful
Lights" },
        { id: 2, image: "https://www.w3schools.com/w3images/nature.jpg", caption: "Nature" },
```

```
    ]);

    const [newImage, setNewImage] = useState("");

    function addFigure() {
        if (newImage) {
            setFigures([...figures,
            {
                id: Date.now(),
                image: newImage,
                caption: "New Image"
            }
            ]);
            setNewImage("");
        }
    }

    const removeFigure = (id) => {
        setFigures(figures.filter(figure => figure.id !== id))
    }

    return (
        <div className="figure-list">
            <input
                type="text"
                placeholder="Image URL"
                value={newImage}
                onChange={(e) => setNewImage(e.target.value)}
            />

            <button onClick={addFigure}>Add Image</button>

            <div className="grid">
                {
                    figures.map(figure => (
                        <BasicFigure
                            image={figure.image}
                            caption={figure.caption}
                            onRemove={() => removeFigure(figure.id)}
                        />
                    ))
                }
            </div>
        </div>
    )
}

export default FigureList;
```

App.css

```
.App {
   text-align: center;
   padding: 20px;
}

.figure-list input {
   margin: 5px;
   padding: 8px;
   width: 180px;
   border: 1px solid #ccc;
   border-radius: 4px;
}

.figure-list button {
   background-color: #4caf50;
   color: white;
   border: none;
   border-radius: 4px;
   padding: 8px 12px;
   cursor: pointer;
   font-size: 12px;
}

   .figure-list button:hover {
      background-color: #45a049;
   }

.grid {

   display: flex;                    Direction → row (left to right)  Items stay in one line.
flex-wrap: wrap;                     If space is not enough,  Items move to the next line.
   gap: 10px;                         Adds spacing between flex items.
   justify-content: center;
   margin-top: 5px;                   adds space above the grid.
}
.basic-figure {
   background: #f9f9f9;
   padding: 10px;
   width: 160px;
}

.figure-image {
   width: 100%;
   border-radius: 6px;
   transition: transform 0.3s ease, box-shadow 0.3s ease;
}

   .figure-image:hover {
```

```
        transform: scale(1.05);
        box-shadow: 0 4px 8px rgba(0, 0, 0, 0.2);
    }

.figure-caption {
    margin-top: 8px;
    font-size: 14px;
    color: #444;
}

.remove-button {
    background-color: #4caf59;
    color: white;
    border: none;
    border-radius: 50%;
    padding: 4px 7px;
    cursor: pointer;
    font-size: 12px;
   }

    .remove-button:hover {
        background-color: #c0392b;
    }
```